

# Measurement Report - Verification of Profile - Demo-Report-Matrix-verify

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## Report Scope

The following profile verification runs are included:

- Demo-Report-Matrix-verify, Instrument: X-Rite ColorMunki · 1 verification run

**No. of Measurements:**

1

**Date range:**

2026-08-10 – 2026-08-10

## How to read this report

This report compares what your instrument measured against the chart's design colours (the reference values the chart was built from). Every number is a colour difference ( $\Delta E_{00}$ ): 0 is a perfect match, 1–2 is barely visible, and 10 or more is clearly different.

- Colour accuracy — the  $\Delta E_{00}$  across the patches, split so you can see the bulk of the chart (all patches and the best 95%) apart from the few hardest patches (the worst 5%). Each metric is judged against your Pass thresholds.
- Paper white & darkest black — the brightest and deepest patches ( $L^*$ ), a quick health check of your paper and maximum ink.
- Cube corners — paper white, composite black and the six primary and secondary inks. These say as much about your inks as about the instrument.

What the numbers mean depends on how the chart was printed:

- A profiling chart is printed **WITHOUT** colour management (the raw print you measure to build a profile). It is not expected to match the design closely, so the  $\Delta E$  can look large — that's normal. Here it is the **CHANGE** between dated reports that matters, not a single value.
- A verification chart is printed **THROUGH** your finished profile — ChromIQ converts the sheet itself and prints it with the printer's colour management off. It **SHOULD** match the design closely, so low  $\Delta E$  and passes mean the profile is still accurate; rising numbers over time tell you when it's worth re-profiling. (Printed raw instead, the same sheet is a printer drift check — the report says which way each sheet was printed.)

Some of a chart's design colours can be brighter or more saturated than this printer and paper can physically produce — no profile can print them, however good it is. Where the report can tell (it asks the run's profile), it splits the colour-accuracy figures into two groups: "Within the profile's gamut" — the colours that were genuinely printable, the fair measure of accuracy — and "Beyond it" — the unreachable ones, whose distance describes the limit of the gamut, not a mistake of the profile. Every patch stays counted and visible; the Pass/Fail verdict judges the within-gamut figures.

Because the design reference never changes, comparing dated reports of the same chart on the same printer is a clean, reliable signal of drift — ageing inks, a wandering printer, or an instrument going off. Save a report after each measurement to build that history. Screen and print colours here are approximate; the numbers come from your measurement file and are exact.

## Report Results

The following results are extracted from the detailed Colour accuracy data shown below for each measurement run. Where a sheet's colours are split into within / beyond the profile's gamut, Pass and Fail judge the within-gamut figures — colours the profile could never print are not counted against it.

| Metric                           | 2026-08-10 |
|----------------------------------|------------|
| Average $\Delta E$ , all patches | Fail       |
| Average $\Delta E$ , lowest 95%  | Fail       |
| Average $\Delta E$ , highest 5%  | Fail       |
| Maximum $\Delta E$ , all patches | Fail       |
| Maximum $\Delta E$ , lowest 95%  | Fail       |

## Detailed data per measurement run

### Measurement run — 2026-08-10T11:00:00 — 105 patches

Profile name: Demo-Report-Matrix-verify

#### How this verification was produced

|                                      |                                                                                                                                        |
|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| What this measured                   | how accurate this profile is — the sheet was the profile's own prediction, made real                                                   |
| Printed                              | through this run's profile · relative colorimetric                                                                                     |
| Colour management at the printer     | off — ChromIQ printed the sheet itself                                                                                                 |
| Profile                              | Demo-Report-Matrix.icc · 2026-08-10                                                                                                    |
| Take care                            | the profile has been rebuilt since this sheet was printed, so these figures describe an older profile than the one now in the run      |
| Readings belong to this chart        | verified — every patch holds the colour the chart asked for                                                                            |
| How the colours were judged          | relative to this sheet's own paper white — the print mapped white to the paper, so the paper itself is not counted against the profile |
| Reference for the $\Delta E$ figures | the chart's design colours                                                                                                             |

#### Colour accuracy ( $\Delta E_{00}$ vs the chart's design)

| Metric                           | Within gamut | Beyond it | All patches | Threshold | Result |
|----------------------------------|--------------|-----------|-------------|-----------|--------|
| Average $\Delta E$ , all patches | 2.51         | 2.18      | 2.49        | 2.00      | Fail   |
| Average $\Delta E$ , lowest 95%  | 2.29         | 2.09      | 2.28        | 2.00      | Fail   |
| Average $\Delta E$ , highest 5%  | 6.65         | 2.77      | 6.65        | 2.00      | Fail   |
| Maximum $\Delta E$ , all patches | 7.26         | 2.77      | 7.26        | 3.00      | Fail   |
| Maximum $\Delta E$ , lowest 95%  | 6.14         | 2.67      | 6.14        | 3.00      | Fail   |
| Spread (std. dev.)               | 1.75         | 0.48      | 1.69        | —         | —      |

97 of this sheet's colours lie within what the profile (Demo-Report-Matrix.icc) can print, 8 beyond it. The Result judges the within-gamut figures: a colour beyond the gamut was never printable, so its distance describes the gamut's limit, not a mistake of the profile — those patches are still shown, in their own column, and their stability over time is a drift signal.

#### Paper white & darkest black

White (4) — L\* 100.0

Black (7) — L\* 0.0

#### Cube corners (the eight ink extremes)

| Corner       | Expected | Measured | $\Delta E_{00}$ |
|--------------|----------|----------|-----------------|
| White (1)    |          |          | 0.26            |
| Black (5)    |          |          | 6.14            |
| Red (11)     |          |          | 1.84            |
| Green (13)   |          |          | 1.61            |
| Blue (10)    |          |          | 6.08            |
| Cyan (12)    |          |          | 0.80            |
| Magenta (14) |          |          | 1.11            |
| Yellow (9)   |          |          | 2.77            |

Worst patches

| Patch | Expected                                                                          | Measured                                                                          | $\Delta E00$ | Patch | Expected                                                                          | Measured                                                                          | $\Delta E00$ |
|-------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------|-------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------|
| 15    |  |  | 7.26         | 49    |  |  | 6.03         |
| 55    |  |  | 7.19         | 39    |  |  | 5.39         |
| 17    |  |  | 6.51         | 57    |  |  | 5.35         |
| 5     |  |  | 6.14         | 104   |  |  | 5.10         |
| 6     |  |  | 6.14         | 103   |  |  | 4.89         |
| 7     |  |  | 6.14         | 31    |  |  | 4.45         |
| 8     |  |  | 6.13         | 56    |  |  | 4.33         |
| 10    |  |  | 6.08         | 61    |  |  | 4.23         |